

EXCALIBUR: A Lifecycle Threat Model for Adaptive CAN Intrusion Detection

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Abstract—Adaptive, online-learning intrusion detection systems (IDS) are gaining traction for the automotive CAN bus. Their calibration phase has not been examined as something an attacker could target. We test that gap with a bounded-budget timestamp-drift attack against a minimal EWMA Z-score online IDS, run on the real ORNL ROAD dataset across three CAN IDs, four ambient drive sessions, and a seven-point budget sweep from 0.01 to 0.5. Attacker budget predicts baseline shift almost exactly, holding at a 0.0050 ratio across all 70 points tested, with just one small deviation, three orders of magnitude below the shift itself. Early-window flag suppression appears wherever clean traffic isn’t already near the measurement floor. Where it is, the effect is too small to see rather than absent, confirmed by overlapping confidence intervals. Over a longer window the effect fades as EWMA adaptation absorbs it. A median/MAD estimator reduces flagging under attack by about one frame out of 200, and this reduction does not scale with attacker budget. The poisoning mechanism itself is not new; our contribution is showing it holds as a calibration-time attack on real CAN traffic.

Index Terms—intrusion detection, controller area network, adaptive systems, data poisoning, automotive security

I. INTRODUCTION

Several recent CAN intrusion detectors learn continuously instead of shipping with fixed parameters. Meta-learning lets a detector adapt to new attack types from only a few examples [1]. Neuro-fuzzy inference and adaptive autoencoders tune themselves to new traffic without full retraining [2], [3], and online change-point detection extends the same idea to message timing [4]. Each of these keeps learning after deployment instead of freezing at design time. Vehicles are also moving toward software-defined architectures with more frequent over-the-air (OTA) updates [5], [6], [7]. That trend is our motivation, not our mechanism. A faster update cadence does not prove that adaptive IDS recalibrate on the same schedule, or at all, in any car currently on the road. What it does show is that events resembling calibration, moments where a learning component is initialized or reset, are turning into a routine part of a vehicle’s software lifecycle instead of a one-time factory step. On that basis alone we think the question is worth asking. We are not claiming the window occurs on a fixed schedule, or that it occurs at all in production.

Our question is narrow: can an attacker with bounded access during that calibration window shift what the detector learns as normal, and does that shift survive long enough to matter? We call the resulting threat model and its case-study

instantiation EXCALIBUR, after the phase it targets. The reference IDS, dataset scope, and statistical evidence below show that the calibration-time mechanism is measurable. They do not establish its severity or prevalence in deployed systems.

This work does not propose a new detection algorithm, and it does not claim adaptive CAN IDS are running in production vehicles. Section II traces the poisoning mechanism itself to prior work outside the automotive domain, so we are not claiming a new attack class either. Online poisoning theory already exists, demonstrated on non-automotive learners, but nobody has treated the calibration window of an automotive CAN IDS as an attack surface in its own right. We name that surface and show, with a case study, that the attack it predicts is measurable on real vehicle data.

Concretely, this paper contributes:

- A lifecycle-aware threat model that treats the calibration window of an adaptive CAN IDS, instead of its steady-state operation, as the attacker-reachable interface (Section III).
- A timestamp-drift attack, run under a bounded attacker budget, against a minimal EWMA/Z-score online CAN IDS. Testing it on the real ORNL ROAD dataset across three CAN IDs and four ambient drive sessions shows that baseline shift grows with both attacker budget and injected drift magnitude and stays stable across sessions, while the attack’s visible effect on early-window flag rate depends on how close clean-traffic flagging already sits to the measurement floor in a given session (Section IV).
- A median/MAD mitigation with a bounded update rate that closes the case study, along with a full accounting of what remains open (Section VI).

II. RELATED WORK

A. Adaptive and Online-Learning CAN IDS

Adaptive CAN intrusion detection is no longer a one-off proposal. Several recent systems independently argue for it. Meta-IDS applies meta-learning (Meta-SGD) to give a CAN detector few-shot adaptability across attack types [1]. Al Isawi et al. build an adaptive neuro-fuzzy inference system for CAN attack detection on electric vehicles, evaluated on real bus traffic [2]. Kim, Im, and Lee propose an adaptive autoencoder IDS with a single, self-tuning threshold for CAN networks [3]. A CUSUM-based change-point detector for CAN message

timing, attributed to Olufowobi et al. and surveyed in a recent evaluation of vehicular IDS, applies the same adaptive-statistical approach to message timing that this paper’s own detector uses [4]. Four separate lines of work treat adaptivity as a core design goal, not an afterthought. None of them examine what happens during the window in which that adaptivity is first calibrated.

B. Why Adaptation Is Becoming More Frequent

Industry and policy sources agree the cadence of vehicle software updates is increasing. Whether that cadence maps onto IDS recalibration is a different question, and one we do not try to settle here. Industry guidance contrasts today’s typical cadence of one or two updates per vehicle per year with the software-defined vehicle (SDV) goal of continuous OTA delivery [5]. The International Energy Agency (IEA)’s independent analysis backs this trajectory, tracing continuous post-sale software evolution of ADAS and battery-management systems back to Tesla’s 2012 introduction of OTA updates [6]. A systematic literature review of SDV ecosystems goes further, calling the traditional practice of tying vehicle updates to mechanical maintenance schedules outdated, and describing more frequent OTA delivery as a lifecycle shift on its own, independent of any mechanical trigger [7]. Regulation reinforces this trend: UN Regulation No. 156 mandates a software update management system for type approval [8], and ISO 24089 provides the corresponding technical process for software update engineering [9]. None of these sources say anything about IDS calibration directly. We cite them only to show that software updates, and any learning components bundled with them, happen more often than they used to.

C. Poisoning and Adversarial Attacks on Automotive IDS

The broader machine-learning literature already shows that online learners can be poisoned during training, not just fooled at inference [10], [11]. Kloft and Laskov examine how injected adversarial data corrupts an online centroid anomaly detector, the closest general-purpose analog to the attack studied here [12]. It is non-automotive, but it demonstrates the mechanism: an attacker corrupting a learner’s state while that state is still forming. Closer to the calibration-time framing here, Korycki and Krawczyk show that streaming classifiers can be fooled by data crafted to look like ordinary concept drift, steering the learner’s adaptation off course [13], the same failure mode this paper’s threat model targets in the CAN domain specifically.

Automotive IDS security research has not followed that thread. Three independent studies target adversarial manipulation of CAN or in-vehicle-network IDS, and all three attack a fixed, already-trained detector at inference time, not during its calibration phase. Cerracchio et al. show that a CAN-bus ML-IDS can be evaded without knowledge of its internals, and propose adversarial training as a countermeasure [14]. A separate study adapts white-, grey-, and black-box evasion methods borrowed from computer vision to CAN’s tabular, real-time setting [15]. Aloraini et al. go further, training a

TABLE I
POSITIONING RELATIVE TO RELATED WORK

Work	Real CAN Data	Attack Phase	Attack Type
Cerracchio et al. [14]	✓	Inference (fixed model)	Evasion
ACM TCPS [15]	✓	Inference (fixed model)	Evasion
PMC IVN [16]	✓	Inference (fixed model)	Evasion
Kloft & Laskov [12]	× (non-auto.)	Calibration	Poisoning
This work	✓	Calibration	Poisoning

substitute model to attack an in-vehicle-network IDS in black-box fashion and dropping its F1 score from 95% to 38% in one setting and 97% to 79% in another [16]. Table I makes the resulting split explicit: the automotive rows all attack a fixed, already-trained model at inference time, while this paper’s row is the only one that uses real CAN data and targets the calibration phase.

Existing automotive IDS security work targets trained detectors, whether attacking, defending, or evaluating them, and does not examine what happens before the detector has learned normal traffic. That gap, not the timestamp-drift mechanism itself, is this paper’s contribution: an instantiation of calibration-time poisoning against an adaptive CAN IDS, evaluated on real vehicle traffic, with the attacker-access assumption stated explicitly instead of left implicit. The attack results below demonstrate that this gap exists and is measurable.

III. THREAT MODEL

A. Adversary

This is a modeling premise, not a demonstrated fact. We assume the attacker has CAN bus access during an IDS calibration window and can perturb a bounded fraction (budget) of frames belonging to a single target CAN ID within that window only. Traffic outside the window and other IDs are untouched. Granting a capability in order to study its consequences is standard threat-modeling practice. Doing so here lets us measure what the capability would achieve if an attacker had it; it does not mean the capability is achievable against a production vehicle today, or that any deployed IDS calibrates on the schedule we describe. Section IIB discusses why calibration-like events may recur. Section VI restates that we have established neither the access path nor the recalibration trigger.

B. Explicit Non-Goals

The threat model does not cover electronic control unit (ECU) functional, sensor, or Unified Diagnostic Services (UDS)/diagnostic calibration, nor physical ECU compromise or tampering with the monitoring point itself. It also does not address whether OEMs should adopt adaptive IDS, or claim that adaptive CAN IDS are currently deployed in production vehicles.

C. Defender

The defender is an online IDS: it computes baseline statistics from the calibration window, then adapts via streaming

per-tick updates. We measure calibration-time impact as *baseline shift*: the change in the learned baseline mean immediately after calibration, read before any post-calibration online update can mask it.

IV. CASE STUDY

A. Reference IDS and Attack

The EXCALIBUR reference IDS is a minimal per-ID online learner (EWMA-updated mean/std, Z-score flagging). It is not intended to be a strong detector. Its role is to make the calibration window measurable. The EWMA decay rate is $\alpha = 0.1$ and the Z-score alert threshold is 3.0.

The attack targets a single CAN ID’s inter-arrival timestamps within the calibration window only. A fixed per-frame drift is added to a bounded fraction of that ID’s frames in the window. Budget denotes this fraction, swept from 0.01 to 0.5 at a fixed drift of 0.005s in the main sweep, and drift itself is separately swept from 0.001s to 0.020s at a fixed budget of 0.2 (Section V). A budget of 0.5 means half of the target ID’s own frames in the calibration window carry the shift, still bounded to one ID and one window, per the threat model (Section III). We do not evaluate whether this budget is achievable by a stealthy attacker in practice.

We report two post-calibration windows throughout: an *early window*, the flag rate measured over the first 200 scored frames of the target ID immediately after calibration ends, and a *full window*, the flag rate measured over the entire post-calibration session. The distinction matters because EWMA adaptation absorbs the injected drift over time, and only the early window reflects the attack’s effect at full strength. Because the early window is fixed at 200 frames, early-window flag rates are necessarily discrete multiples of $1/200 = 0.005$, so we report raw flagged-frame counts alongside rates throughout Section V.

B. Dataset

We use the real ORNL Automotive Dynamometer (ROAD) dataset [17] over HCRL Car-Hacking [18]/Survival Analysis [19] (considered but not evaluated here) and OTIDS/CICIoV2024 (rejected as insufficiently automotive-CAN-specific or too synthetic). ROAD provides real vehicle traffic across multiple ambient drive sessions with labeled attack sessions, supporting a session-time split into calibration and post-calibration segments. ROAD does not natively encode IDS calibration/lifecycle metadata, and no public dataset does. The calibration/post-calibration boundary is synthesized via a session-time cutoff we impose (Section VI).

We evaluate three CAN IDs, 125, 354, and 193, across four ambient drive sessions from the ROAD dataset’s CSV files: `ambient_dyno_drive_basic_long` (`basic_long`), `ambient_highway_street_driving_long` (`highway_long`), `ambient_dyno_drive_winter` (`winter`), and `ambient_dyno_drive_extended_long` (`extended_long`). We use the short forms in parentheses throughout the rest of the paper. Each session uses a 30-second calibration window. Coverage is not uniform:

TABLE II
CASE STUDY RESULTS, TARGET ID 125, SESSION BASIC_LONG
(DRIFT=0.005s)

Metric	Value
Baseline shift at budget 0.01	0.000050
Baseline shift at budget 0.05	0.000250
Baseline shift at budget 0.50	0.002501
Clean early-window flags (of 200)	7 (0.035)
Attacked early-window flags at budget 0.01	4 (0.020)
Attacked early-window flags at budget 0.02	2 (0.010)
Attacked early-window flags at budget ≥ 0.05	1 (0.005)
Clean full-window flag rate	0.026577
Attacked full-window flag rate	0.026519–0.026552
Mitigated (RobustIDS) full-window flag rate	0.001916

IDs 125 and 354 do not appear in the `extended_long` session’s calibration window and are therefore evaluated across the remaining three sessions only, while ID 193 is present and evaluated in all four. In total we ran $3 \times 7 + 3 \times 7 + 4 \times 7 = 70$ budget-sweep points across sessions and targets.

C. Mitigation

We call the mitigation RobustIDS. It bundles three changes; we did not isolate a single fix. First, it replaces the EWMA/Z-score estimator with a median/Median Absolute Deviation (MAD) estimator [20]. Second, it adds a bounded per-update delta capping how far any single tick can move the baseline. Third, it raises the alert threshold from 3.0 to 5.0, alongside a variance floor that keeps the MAD-based denominator from collapsing on low-jitter signals. All three are evaluated together in Section VI, so those results describe RobustIDS as a bundle. They cannot say whether the observed effect comes from the estimator swap, the update cap, or the threshold change (Section VI traces how the floor and threshold were arrived at).

V. EVALUATION

Table II reports results for target ID 125 in the `basic_long` session, drift = 0.005s, budgets 0.01–0.5, over the 30s calibration window. Table III reports raw early-window flagged-frame counts (out of 200) for all three targets across all four sessions, at budget 0.01 and budget 0.50, alongside each session’s clean-traffic count. Table IV reports the drift-magnitude sweep at fixed budget 0.2.

A. Baseline Shift Scales With Budget and Drift, and Is Session-Invariant

For target 125 at fixed drift 0.005s, baseline shift rises from 0.000050 at budget 0.01 to 0.002501 at budget 0.50 (Fig. 1). The ratio of shift to budget is 0.005 at every one of these seven points, matching the per-frame drift magnitude exactly. This ratio is not specific to one session: recomputing baseline shift for target 125 at budget 0.50 gives 0.002501 in all three sessions in which it is evaluated (`basic_long`, `highway_long`, `winter`), and target 354 gives the same value, 0.002501, in all three of its sessions. Target 193, evaluated across all four sessions, gives 0.002502 uniformly.

TABLE III

EARLY-WINDOW FLAGGED-FRAME COUNTS (OF 200) WITH 95% WILSON SCORE CONFIDENCE INTERVALS, CLEAN VS. ATTACKED AT BUDGET 0.01 AND BUDGET 0.50 (DRIFT=0.005). BOLD MARKS FLOOR-LEVEL SESSIONS (CLEAN COUNT 0–1) WHERE CLEAN AND ATTACKED INTERVALS OVERLAP ALMOST ENTIRELY.

Target ID	Session	Clean k (95% CI)	Attacked @ 0.01 k (95% CI)	Attacked @ 0.50 k (95% CI)
125	basic_long	7 (0.017, 0.071)	4 (0.008, 0.050)	1 (0.001, 0.028)
125	highway_long	10 (0.027, 0.090)	6 (0.014, 0.064)	1 (0.001, 0.028)
125	winter	0 (0.000, 0.019)	1 (0.001, 0.028)	1 (0.001, 0.028)
354	basic_long	5 (0.011, 0.057)	4 (0.008, 0.050)	1 (0.001, 0.028)
354	highway_long	3 (0.005, 0.043)	4 (0.008, 0.050)	1 (0.001, 0.028)
354	winter	14 (0.042, 0.114)	6 (0.014, 0.064)	1 (0.001, 0.028)
193	basic_long	11 (0.031, 0.096)	9 (0.024, 0.083)	1 (0.001, 0.028)
193	highway_long	0 (0.000, 0.019)	1 (0.001, 0.028)	1 (0.001, 0.028)
193	extended_long	1 (0.001, 0.028)	1 (0.001, 0.028)	3 (0.005, 0.043)
193	winter	16 (0.050, 0.126)	12 (0.035, 0.102)	3 (0.005, 0.043)

TABLE IV

BASELINE SHIFT UNDER THE DRIFT-MAGNITUDE SWEEP, TARGET 125, SESSION BASIC_LONG, BUDGET=0.2

Drift (s)	Baseline shift	Shift/Drift ratio
0.001	0.000200	0.200
0.005	0.001000	0.200
0.010	0.002001	0.200
0.020	0.004001	0.200

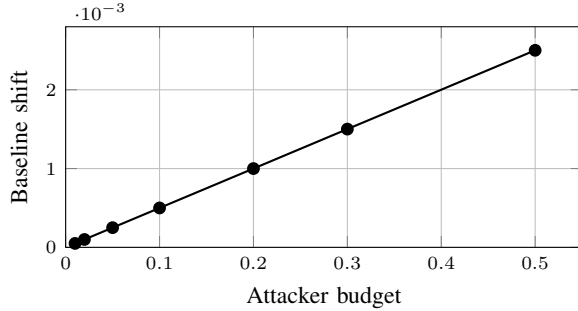


Fig. 1. Calibration-time baseline shift vs. attacker budget, target ID 125, session basic_long, drift 0.005s. Shift/budget is constant at 0.005 across the swept range and across every other session/target pair tested.

Baseline shift is the most robust result in this study for that reason: budget and drift magnitude govern it almost entirely, with negligible session-to-session variation, unlike early-window flag rate (Section VB). Separately, at fixed budget 0.2, sweeping drift from 0.001s to 0.020s gives a constant shift/drift ratio of 0.200 across all four drift levels, equal to the fixed budget. Both sweeps track a linear relationship, baseline shift = budget $\times$ drift, over the ranges tested in each dimension. The joint budget–drift space has not been swept.

Our primary inferential model is a mixed-effects fit with a random intercept grouped by session, pooled across all 70 (session, target, budget) points. It gives a fixed budget coefficient of 0.0050 ($p < 0.001$) and places the between-session variance at the boundary of the parameter space, i.e. too small

for the model to resolve as anything other than negligible. A simpler pooled OLS fit, ignoring session grouping entirely, gives the same coefficient (0.0050, $p < 0.001$). This model also reports $R^2 = 1.000$. That number should not be read as a finding. Baseline shift is computed directly as budget times a fixed drift constant, so a perfect fit is built into the sweep rather than discovered by it. We report the value for completeness only. Adding session as a categorical term to the OLS fit improves it further (ANOVA F -test, $p = 3.6 \times 10^{-7}$), but the improvement is not spread evenly: only the `extended_long` coefficient differs from the `basic_long` reference level (-1.239×10^{-6} , $p < 0.001$), while `highway_long` and `winter` are indistinguishable from it. A `budget` $\times$ `session` interaction is likewise significant ($p = 7.8 \times 10^{-5}$) and again confined to `extended_long`. We read this as a real but small effect rather than a contradiction of session-invariance: the `extended_long` deviation is roughly three orders of magnitude smaller than the shift values it is attached to (10^{-6} against shifts of 10^{-4} – 10^{-3}), and `extended_long` is also the one session in which only a single target (193) is evaluated, so the deviation may reflect that target rather than the session itself. The two are confounded here and cannot be separated with the sessions available.

B. Early-Window Flag Rate Is Conditional on Clean-Traffic Noise Level

Table III reports raw flagged-frame counts across all ten evaluable (target, session) pairs. Two distinct regimes appear, separated by how many clean-traffic frames are flagged in a given session before any attack is applied.

Regime 1: non-floor clean traffic. In seven of the ten pairs, clean-traffic flagging is not near zero (clean counts of 3–16 out of 200). In every one of these seven, the attack suppresses the early-window count monotonically as budget increases: target 125/basic_long’s count drops from 7 to 4 to 1, target 193/winter’s drops from 16 to 12 to 3, with a similar pattern across the remaining pairs. The dose-response pattern holds wherever clean traffic gives the attack room to suppress.

Regime 2: floor-level clean traffic. In the remaining three pairs (target 125/winter, target 193/highway_long, and target 193/extended_long), clean-traffic flagging is already at or within one count of the measurement floor (0 or 1 out of 200). Here the attacked count equals or sits slightly above the clean count at every tested budget (e.g., target 125/winter: clean 0, attacked 1 at every budget from 0.01 to 0.50, and target 193/highway_long: clean 0, attacked 1 at every budget). The 95% Wilson score confidence intervals on these counts (Table III) overlap almost completely between clean and attacked conditions in all three floor pairs (e.g., target 125/winter: clean [0.000, 0.019] vs. attacked [0.001, 0.028]). We cannot say from this data whether the attack has any effect at all in these three pairs. The apparent one-count rise is equally consistent with zero effect, and we do not treat it as a demonstrated result in either direction. Contrast the non-floor pairs. There, clean and attacked intervals separate substantially or entirely by budget 0.50 (e.g., target 193/winter: clean [0.050, 0.126] vs. attacked [0.005, 0.043], non-overlapping). We report the floor pairs instead of dropping them, but we flag them as unresolved. With only 200 scored frames and a clean rate already at the floor, this study’s design cannot tell a genuine small effect apart from no effect at all. Answering that would need a larger early window.

The attack does not universally suppress early-window flag rate at every budget across all sessions. It suppresses reliably when clean traffic isn’t already at the detection floor. At the floor, its effect is undetermined by this study, not merely small. That points to a defender heuristic, stated cautiously: calibration-time poisoning may be hardest to catch on exactly the CAN IDs with the fewest natural anomalies, since those are the ones with the smallest early window to work with. We cannot confirm this from the floor-level pairs alone. Measuring it would take a larger early-window sample (Section VI).

Full-window flag rates remain close between clean and attacked conditions in every session/target pair we examined in detail (e.g., target 125/basic_long: 0.026577 clean vs. 0.026519–0.026552 attacked across the budget sweep), consistent with EWMA adaptation absorbing the calibration-time shift given enough post-calibration ticks, regardless of session or target.

C. Statistical Significance

As a supporting check on the mixed-effects and OLS results above, a paired comparison of clean vs. attacked calibration-window means across the target-125, session-basic_long budget sweep (seven points, 0.01–0.5) gives $t = -2.480$, $p = 0.048$, and a 95% bootstrap confidence interval on baseline shift of [0.000279, 0.001501]. The interval excludes zero, but that is unsurprising given how the attack is built. Budget and drift are set directly, and baseline shift follows from them by design, so this test mostly confirms the simulation is internally consistent. It does not independently validate a real-world effect. We include it anyway for two reasons: it is the only significance test run on real, non-synthetic traffic in this study, and it agrees with the pooled models, which reproduce the

same deterministic relationship to four decimal places across every session and target evaluated.

D. Estimator Comparison: EWMA vs. Fixed-Window

Replacing the EWMA baseline with a fixed-window mean estimator (same calibration data, no mitigation applied) changes the early-window flag-rate gap between clean and attacked traffic, and this comparison is itself session-dependent. On target 125/basic_long, the fixed-window estimator’s early-window count is flat at 4/200 across every tested budget, while the EWMA estimator’s count descends from 4/200 to 1/200 as budget increases. This has only been examined on one session/target pair. Raw-count breakdowns across all ten pairs are left to future work, so we stop short of a directional claim (e.g., “estimator choice flips the sign of the effect”) without that broader check.

E. Mitigation Effect

On target 125/basic_long, RobustIDS’s full-window flag rate on attacked traffic (0.001916) sits well below both the EWMA clean baseline (0.026577) and the raw EWMA attacked rate (≈ 0.02652). This full-window comparison conflates the mitigation’s effect with the estimator/threshold swap itself. To isolate the mitigation’s response to the attack, we additionally ran RobustIDS on *unpoisoned* calibration data across all ten (target, session) pairs, holding the estimator fixed and varying only the calibration traffic. Even so, this only isolates the traffic condition. The estimator swap, the bounded update rate, and the raised threshold stay bundled together throughout, so what follows describes how the bundle behaves, not what any single component contributes.

Under this same-estimator comparison, RobustIDS’s early-window count on clean calibration data is 0/200 in nine of the ten pairs and 1/200 in the tenth (target 193, session extended_long). It never exceeds 1/200 on clean traffic anywhere. Under attack, RobustIDS’s early-window count rises to a flat 1/200 across nearly every budget in every pair, with an isolated 2/200 at the lowest tested budget (0.01) for target 193 in three sessions (basic_long, highway_long, winter), dropping back to 1/200 at every higher budget in those same pairs. The net effect of the attack under RobustIDS is a single additional flagged frame out of 200, consistent across budget, target, and session, rather than a budget-responsive trend: RobustIDS does not show the dose-response pattern that EWMA shows in the non-floor sessions (Section VB), nor does its response scale with attacker budget the way baseline shift does (Section VA). RobustIDS starts and stays near the 200-frame measurement floor under calibration-time poisoning. Given that floor, the small, flat uptick it shows could be a floor artifact instead of a genuine mitigation-resistant residual, and we cannot rule that out here (Section VI). A larger early window is needed before this bundled result can be read either way.

VI. LIMITATIONS

- 1) **The mitigation effect is bundled, and it does not respond to budget.** RobustIDS combines an estimator

swap, a bounded update rate, and a raised alert threshold. Section IVC and Section VF evaluate it as a single bundle throughout; no experiment here isolates which component drives the observed one-count rise. A same-traffic comparison of RobustIDS on clean versus attacked calibration data shows clean-traffic early-window counts holding at 0–1/200, with attacked counts rising to a flat 1–2/200 across every budget. That rise does not scale with budget the way EWMA’s suppression does in non-floor sessions, and we cannot separate it from a single-count floor effect (Item 2).

- 2) **Early-window counts are small, and the floor-level results are unresolved rather than just noisy.** Flag counts are fixed at 200 scored frames and range from 0 to 16 across sessions and targets, which is enough to see a monotonic dose-response trend where clean counts aren’t near zero. Three pairs start with clean flagging at 0–1, though, and there this study’s design cannot tell a real small effect from no effect. The confidence intervals overlap almost completely. We report these three as open questions, not as evidence for or against suppression.
- 3) **Coverage and sample size are limited.** IDs 125 and 354 are absent from the `extended_long` session (Section IV), so only ID 193 has full four-session coverage, and this is also why the small session-level deviation in Section VA cannot be separated from a target-193-specific effect. The significance test and estimator comparison draw on a single session/target pair (target 125, `basic_long`). Extending both across the remaining nine pairs and the drift range in Table IV is future work.
- 4) **Attacker access is a threat-model assumption, and the calibration boundary is a cutoff we imposed.** Section IIIA states this directly: we assume CAN bus access during calibration in order to study what that access would enable. We are not claiming it is achievable in a production vehicle, and we have no evidence any real attacker has obtained it. The calibration/post-calibration split works the same way. It is a session-time cutoff applied to ROAD data by us, not a boundary the dataset or any deployed system defines.
- 5) **The lifecycle-trigger scenario stays unresolved.** Section IIB shows that OTA update cadence is increasing. It does not show that this cadence produces IDS recalibration events, on any schedule, in any deployed system. Public literature does not confirm whether recalibration ties to factory, dealer, or field-replacement events. We treat the OTA trend as motivation for asking the question, not as a load-bearing premise. The poisoning mechanism itself holds no matter what triggers recalibration in practice.
- 6) **Two measurement bugs were caught during implementation.** An early pipeline version read IDS state after post-calibration scoring, so millions of EWMA updates had already averaged the calibration-time corruption away before it was measured. The fix snapshots

state immediately after calibration, the convention used throughout this paper. An early RobustIDS version also flagged 99.4% of clean traffic because its MAD-based denominator collapsed on low-jitter signals, fixed with a variance floor and a threshold raised from 3.0 to 5.0 (Section IV). Both bugs came from measuring adaptive-system state at the wrong point in its update cycle.

- 7) **This is not a new attack class.** Training-time poisoning of online learners is established in the machine-learning literature [12]. What’s new here is the automotive instantiation targeting adaptive CAN IDS calibration windows.

VII. CONCLUSION

This paper names the calibration window as an attacker-reachable surface and backs that claim with a case study on real vehicle data across three CAN IDs and four ambient drive sessions. Baseline shift is the clearest result. It scales linearly with both attacker budget and drift magnitude, and it holds steady across sessions and targets: a mixed-effects model pooled across all 70 evaluated points gives a single budget coefficient of 0.0050, with only one session showing a detectable deviation, three orders of magnitude smaller than the shifts it sits next to. Part of that linearity comes from how the attack itself is constructed, so we present it as a consistency check on the simulation as much as an empirical finding. Early-window flag-rate suppression is messier. It shows up wherever clean-traffic flagging isn’t already near the measurement floor. Where it is, we cannot say whether the effect is present or absent. The original single-session analysis could not draw that distinction either.

A same-traffic ablation shows the bundled median/MAD mitigation reduces early-window flagging relative to an unmitigated attack. We did not evaluate the estimator swap, bounded update rate, and raised threshold that make up RobustIDS separately, so this result describes the bundle, not any one design choice inside it. The effect is small: roughly one additional flagged frame out of 200 under attack, against 0–1 on clean traffic, and it does not scale with attacker budget. We also cannot fully separate it from the same 200-frame floor phenomenon that leaves several other findings in this study unresolved. Taken together, these results support treating the calibration window as an attacker-reachable phase worth explicit consideration when designing adaptive CAN IDS. Read them as a threat-modeling exercise on real traffic, not as a demonstrated production attack. There is a methodological lesson here too: small-sample early-window metrics need raw-count reporting and same-traffic ablations, not rate comparisons alone, and some floor-level questions in this design simply need a bigger window before anyone can answer them.

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